Key

**Quiz 1**

Instructor: Dr. Faisal Bukhari — Marks: 50

Probability & Statistics — Time allocated: 35 mins

**Name:** — **Roll No:** BSDSF22M022

**Attempt all Questions**

**1.** Identify the Appropriate Level of Measurement (Nominal,Ordinal, Interval, Ratio) and Give Reasoning (in one line) **[15 points]**

**a)** Marital status:

Measurement: Nominal

Reasoning: Can be categorised (single, married) but there is no inherent order

**b)** Weight of a new born baby:

Measurement: Ratio

Reasoning: value multiple of another (a baby person who weighs 10 pouds weighs twice as much as baby who weigh 5 pounds)

**c)** Job classifications of employees within an organization:

Measurement: Ordinal

Reasoning: canbe ranked (entry-level job, senior-level)

**d) Temperature of Patient:**

**Measurement:** Interval

**Reasoning:** One data value is not multiple of another

**e) IQ score of students:**

**Measurement:** Interval

**Reasoning:** There is no zero point for IQ

**2.** Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated

(a) if each couple is to sit together? **[6 points]**

Four couple canbe arranged in 4! ways = 24

Each pair has 2! ways = 2! x2! x2! x2!

Total no of way = 4! x2! x2! x2! x2!

= 384 ways

(b) if all the men sit together to the right of all the women?

No of arrangement of 4 men = 4!

No of arrangement of 4 women = 4!

= 24x24

= 576

**3.** On February 1, 2003 the Space Shuttle Columbia exploded. This was the second disaster in 113 space missions for NASA. On the basis of this information, what is the probability that a future mission is successfully completed? **[5 points]**

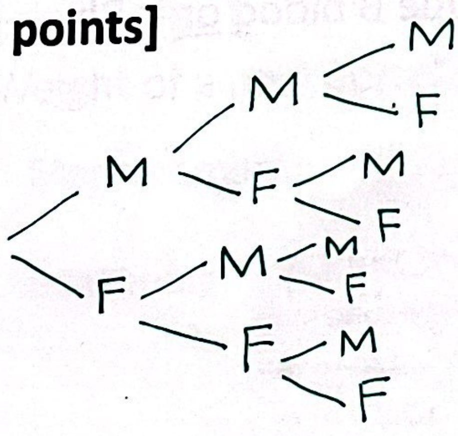
Successful mission = 113 - 2 = 111

$= \frac{111}{113} = 0.9823$

98.23 % probability --

**4.** An experiment consists of determining the order of birth for three children in a family, focusing on their gender. Each child can be either male (M) or female (F). Construct a tree diagram to show the different combinations of genders for the three children and list the outcomes in the sample space S. Assume the probability of having a male child (M) and a female child (F) is each 1/2 . State Probability of each outcome.

**[10 points]**




MMM : $P = \frac{1}{8}$

MMF : $P = \frac{1}{8}$

MFM : $P = \frac{1}{8}$

MFF : $P = \frac{1}{8}$

FMM : $P = \frac{1}{8}$

FMF : $P = \frac{1}{8}$

FFM : $P = \frac{1}{8}$

FFF : $P = \frac{1}{8}$.

**5.** A blood bank catalogs the types of blood, including positive or negative Rh-factor, given by donors during the last five days. The number or donors who gave each blood type is shown in the table. A donor is selected at random. **[14 points]**

| | | Blood type | | | | |
|---|---|---|---|---|---|---|
| | | O | A | B | AB | Total |
| Rh-factor | Positive | 156 | 139 | 37 | 12 | 344 |
| | Negative | 28 | 25 | 8 | 4 | 65 |
| | Total | 184 | 164 | 45 | 16 | 409 |

1.Find the probability that the donor has type O or type A blood.

$P(O) = 184/409$

$P(A) = 164/409$

$= P(O) + P(A)$

$= \frac{184}{409} + \frac{164}{409} = 0.8509$ $\Rightarrow 0.44 + 0.40$

85%

2. Find the probability that the donor has type B blood $\boxed{\text{or}}$ is Rh negative. $P(B) + P(N) - P(B \cap N)$

$\frac{45}{409} + \frac{65}{409} - \frac{8}{409} = \frac{102}{409} = 0.249$

24%